





► **Proof:** We will solve this using Lagrange multipliers. Consider the Lagrangian:

$$\mathcal{L}(\pi) = \pi \left[\frac{d\pi}{d\eta} \right] - \alpha(\pi[1] - 1) - \beta(\pi[U] - u)$$

- Take the functional derivative with respect to π and set to zero

$$\frac{\partial \mathcal{L}}{\partial \pi} = \log \frac{d\pi}{d\eta} - 1 - \alpha - \beta U = 0$$

► Note that since $\log Z(\beta)$ is the cumulant generating function of $\mathcal{U}$ we have

$$\frac{\mathrm{d}}{\mathrm{d}\beta}\langle U\rangle_{\beta} = -\frac{\mathrm{d}^2}{\mathrm{d}\beta^2}\log Z(\beta) = -I(\beta)$$

► Therefore $\langle U \rangle_\beta$ is a strictly decreasing function of β

► Solving for π we obtain

$$\pi(x) = \eta(x) \exp(-\beta U(x) - 1 - \alpha)$$

$$\alpha \pi_{\beta}(x)$$

$$\langle U \rangle_{\beta} = \frac{\mathrm{d}}{\mathrm{d}\beta} \log Z(\beta)$$

► By the intermediate value theorem we have there exists a unique β such that $\langle U \rangle_\beta = u$

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- Take the functional derivative with respect to π and set to zero

$$\frac{\partial \mathcal{L}}{\partial \pi} = \log \frac{d\pi}{d\eta} - 1 - \alpha - \beta U = 0$$

- Solving for π we obtain

$$\pi(x) = \eta(x) \exp(-\beta U(x) - 1 - \alpha) \propto \pi_\beta(x)$$

- Note that since $\log Z(\beta)$ is the cumulant generating function of U we have

$$\langle U \rangle_\beta = \frac{d}{d\beta} \log Z(\beta) \qquad \frac{d}{d\beta} \langle U \rangle_\beta = - \frac{d^2}{d\beta^2} \log Z(\beta) = - I(\beta)$$

- Therefore $\langle U \rangle_\beta$ is a strictly decreasing function of β
- By the intermediate value theorem we have there exists a unique β such that $\langle U \rangle_\beta = u$

ANNEALING PARAMETER